

Two local branches and the exact conductor quotient

Split-Zero programme: coordinate calculation

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Result SZ-20260921-008. This reader gives the complete RLA1–35 and RQT1–22 proofs. The original certified point and directional proofs are linked at their exact public editions and included in the accompanying cumulative source and source archive. All claims concern the stated geometric family. The notation α, β denotes its two spectral roots; the original unit scale and quartet constant are explicitly distinguished below.

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1 The complete local two-branch object and the original finite conductor

This is a derivation for the certified member of the original geometric family, with its original parameters recovered by exact inverse maps. The point and full-series bounds are RPZ1–19 [9]; the conjugation, old four-label return and finite reciprocal calculation are RPCJ1–23, RPE1–26 and RPAC1–15 [5, 8, 4]. The companion complete directional source, RPD1–31 including RPD12a [6, 7], supplies the exact nonzero real Jacobian used below. Its identity RPD4 also follows directly from RPZ10–15, as recalled here. The original matrix and Gamma frames are WCF1–6 [10], whose complete pinned proof retains the Meixner–Pollaczek provenance, DLMF (18.19.8)–(18.19.9), (18.22.8), (18.23.7). The original factor family is PCL1–10 [3], with its complete pinned proof. We prove the local constructions and all finite module identities below; elementary analytic inversion, polynomial division and matrix elimination are used with their proofs, not claimed as new general theorems. The four marked states retain the canonical reader by The Clankers, its theorem **explicit-four-time-Jacobian-collision**, and its distinction between the antecedent three-dimensional example and the four-dimensional marked construction [2]. The certified interval arithmetic used in RPZ is Fredrik Johansson’s Arb [11]; the classical Gamma dictionary is [1]. No arithmetic identification is made.

1.1 Original functions, coefficient reflection and the exact parameter inverse

Let $p = (z, x)$ and let $p_* = (z_*, x_*)$ be the unique real root in the exact rational rectangle RPZ4. Retain $u = 1/z$, the four primary exponents

$$(\rho_M^{\text{prim}}, \rho_N^{\text{prim}}, \rho_T^{\text{prim}}, \rho_E^{\text{prim}}) = (3/4 + 3i, 3/4 - 3i, 1/4 + 3i, 1/4 - 3i), \quad (\text{RLA1})$$

$$\delta = 1/4, \quad \gamma = 3, \quad \beta_0 = 143/8, \quad \eta = 21025/256.$$

The subscript on β_0 distinguishes this unchanged original quartic constant from the spectral root $\beta(p)$ constructed below. Likewise α_{unit} denotes the unchanged nonzero real scale written α in RPZ3: the actual unit remains $a = \alpha_{\text{unit}}(1 + ix)$. Put $w_a = \rho_a^{\text{prim}} - 1/2$, $V_{ar} = w_a^r$, $J_d = \text{diag}(1, -1, -1, 1)$, $W(x) = I + ixJ_d$, $\zeta = e^{2\pi i/5}$ and $D_j = \text{diag}(\zeta^{-ja})_{a=1}^4$. The complete original functions, with $1 \leq a \leq 4$, $0 \leq r \leq 3$, are

$$\begin{aligned} \mathcal{R}(z) &= \sum_{n \geq 0} R_n z^n, \\ (R_n)_{ar} &= \sum_{\substack{h, k \geq 0 \\ 2h+4k=5n+r+1-a}} \frac{(\beta_0/3)^h \eta^k (-5)^{h+k-n} (a/5)_{h+k-n}}{h!k!}, \\ M_j(p) &= W(x)^{-1} V \mathcal{R}(z)^{-1} D_j \mathcal{R}(z) V^{-1} W(x), \\ f_j(\xi, p) &= Q_0(M_j(p)(e^{\rho_a^{\text{prim}} \xi})_a), \\ Q_0(y) &= y_1 y_4 - y_2 y_3, \\ G(\xi, p) &= f_1(\xi, p) f_2(\xi, p), \\ G^\#(\xi, p) &= \overline{G(\bar{\xi}, \bar{p})}, \\ E_A(\xi, p) &= G(\xi, p) G^\#(\xi, p), \\ g(t, p) &= e^{-4\omega(t)} E_A(\omega(t), p), \\ \omega(t) &= -i \arctan t. \end{aligned} \quad (\text{RLA2})$$

Empty sums are zero. RPZ5–9 prove convergence, holomorphicity and matrix invertibility on a complex neighborhood of p_* after shrinking it. This last assertion also follows directly from the entire convergence bound and nonzero determinants at p_* . The reflection in RLA2 conjugates

the Taylor coefficients in all three variables. It is holomorphic; it is not pointwise conjugation at a complex parameter. The real-coefficient series $\mathcal{R}$, $\bar{V} = J_c V$, $\overline{W(\bar{x})} = J_c W(x) J_c$, and $Q_0(J_c y) = -Q_0(y)$ give $f_{5-j} = -f_j^\#$ identically. Here J_c exchanges indices 1, 2 and 3, 4. Thus $G^\# = f_4 f_3$ and the two minus signs cancel in E_A . This proves the factor identity throughout the complex neighborhood.

All following derivatives without displayed arguments are at $(0, p_*)$. Set $b_* = f_2 \neq 0$, $B = G_\xi \neq 0$, $A_z = G_z$, $A_x = G_x$, $b = |B|^2 > 0$ and

$$\begin{aligned} \Delta &= \Im(\overline{A_z} A_x) \neq 0, & \mathcal{H}(p) &= \begin{pmatrix} G(0, p) \\ G^\#(0, p) \end{pmatrix}, \\ J_{\mathbb{C}} &= D\mathcal{H}(p_*) = \begin{pmatrix} A_z & A_x \\ \overline{A_z} & \overline{A_x} \end{pmatrix}, & \det J_{\mathbb{C}} &= -2i\Delta. \end{aligned} \tag{RLA3}$$

The nonvanishing statement is proved, rather than assumed: RPZ12–13 give $\|I - PJ\| < 1$ for the full real Jacobian $J = D_p(\Re f_1, \Im f_1)$ and RPZ11 gives $\det P \neq 0$. The convergent geometric matrix series in $I - PJ$ inverts PJ , so J is invertible. At the zero, multiplication by b_* takes J to $D_p(\Re G, \Im G)$ and has determinant $|b_*|^2$. Its determinant is precisely Δ , proving RLA3. The quantitative enclosure RPD31 has $\Delta < 0$, but nonvanishing alone suffices for the construction.

There is an actual holomorphic inverse $p = \mathcal{P}(r_+, r_-)$ to $\mathcal{H}$, with

$$J_{\mathbb{C}}^{-1} = \frac{1}{-2i\Delta} \begin{pmatrix} \overline{A_x} & -A_x \\ -\overline{A_z} & A_z \end{pmatrix}, \quad p_{n+1} = p_n - J_{\mathbb{C}}^{-1}(\mathcal{H}(p_n) - (r_+, r_-)^\top), \quad p_0 = p_*. \tag{RLA4}$$

To prove it, choose a closed complex ball about p_* on which $\|I - J_{\mathbb{C}}^{-1} D\mathcal{H}\| \leq 1/2$; continuity and RLA3 give such a ball. Restrict (r_+, r_-) so the displacement of its center under RLA4 is at most one quarter of that ball's radius. Integration along a segment gives a contraction ratio 1/2 and an image inside three quarters of the radius. The differences of successive iterates are bounded by a geometric series. The uniform limit exists and is the unique fixed point, hence exactly a solution of $\mathcal{H}(p) = (r_+, r_-)$. Every iterate is holomorphic in the parameters; local uniform convergence and the coefficientwise Cauchy integral formula show that the limit is holomorphic. Uniqueness proves both inverse compositions after restriction to these neighborhoods. In particular all original coordinates are recovered, with $u = 1/\mathcal{P}_z$ and $a = \alpha_{\text{unit}}(1 + i\mathcal{P}_x)$, without identifying a scale or an original parameter with a chosen constant.

Since $g(0, p) = r_+ r_-$, its zero divisor is the union of the two smooth curves $r_+ = 0$ and $r_- = 0$, meeting transversely only at p_* . Each factor has multiplicity one. On the real parameter slice $r_- = \overline{r_+}$, so their union meets that slice only at p_* . This is the complete local zero divisor of the original zeroth moment, not only its tangent cone.

1.2 The two actual spectral roots and their full analytic units

Put $F_+(t, p) = G(\omega(t), p)$ and $F_-(t, p) = G^\#(\omega(t), p)$. Their t derivatives at the base point are $-iB$ and $-i\bar{B}$, respectively, both nonzero. There are unique holomorphic roots $\alpha(p), \beta(p)$ close to zero, given by the convergent iterations

$$\begin{aligned} t_{n+1}^+ &= t_n^+ - F_+(t_n^+, p)/(-iB), & t_0^+ &= 0, \\ t_{n+1}^- &= t_n^- - F_-(t_n^-, p)/(-i\bar{B}), & t_0^- &= 0, \\ \alpha(p) &= \lim_n t_n^+, & \beta(p) &= \lim_n t_n^-. \end{aligned} \tag{RLA5}$$

The proof is the one-dimensional version of RLA4: on a sufficiently small fixed complex disk and parameter neighborhood each derivative of the iteration in its first variable has modulus at most 1/2, and its value at zero has modulus at most one quarter of the radius. The same geometric and Cauchy integral arguments give unique holomorphic limits. Consequently these

are roots of the actual infinite-series factors. Because f_2 and f_3 stay nonzero, they are exactly the roots of the original f_1 and f_4 , respectively. The original spectral coordinates are $\xi_+ = \omega(\alpha)$ and $\xi_- = \omega(\beta)$; the inverse spectral morphism is $t = \tan(i\xi) = i \tanh \xi$ on this neighborhood. Multiplication of its derivative by $\omega'(t)$ and the values at zero, or the defining arctangent identity, verifies both compositions.

The complete units and complete factorization are

$$\begin{aligned} U_+(t, p) &= \int_0^1 \partial_t F_+(\alpha(p) + \theta(t - \alpha(p)), p) d\theta, \\ U_-(t, p) &= \int_0^1 \partial_t F_-(\beta(p) + \theta(t - \beta(p)), p) d\theta, \\ F_+ &= (t - \alpha)U_+, & F_- &= (t - \beta)U_-, \\ U(t, p) &= e^{-4\omega(t)}U_+(t, p)U_-(t, p), & q(t, p) &= (t - \alpha)(t - \beta), \\ g(t, p) &= U(t, p)q(t, p), & U(0, p_*) &= -|B|^2 = -b. \end{aligned} \tag{RLA6}$$

Integration of the derivative proves both divisions exactly. Integration of locally convergent Taylor series proves the units holomorphic; their values at the base point are $-iB$ and $-i\bar{B}$. They, their product, and the center exponential are nonzero on a smaller neighborhood. No coefficient of any unit has been removed. In particular

$$r_+ = -\alpha U_+(0, p), \quad r_- = -\beta U_-(0, p), \quad g_0 = U(0, p)\alpha\beta. \tag{RLA7}$$

Thus $\alpha = 0$ and $\beta = 0$ are precisely the two original parameter branches, with the original $r_{\pm}$ factors retained.

The roots themselves are a pair of holomorphic coordinates. Implicit differentiation of the two proved zero equations gives

$$D\alpha(p_*) = -\frac{i}{B}(A_z, A_x), \quad D\beta(p_*) = -\frac{i}{\bar{B}}(\bar{A}_z, \bar{A}_x), \quad \det D(\alpha, \beta)(p_*) = \frac{2i\Delta}{b} \neq 0. \tag{RLA8}$$

An exact inverse, not just the derivative, is constructed as follows. For proposed roots (A_0, B_0) close to zero iterate

$$p_{n+1} = p_n - J_{\mathbb{C}}^{-1} \begin{pmatrix} F_+(A_0, p_n) \\ F_-(B_0, p_n) \end{pmatrix}, \quad p_0 = p_*. \tag{RLA9}$$

Its parameter derivative vanishes at $(A_0, B_0, p) = (0, 0, p_*)$. The ball argument of RLA4 proves a unique holomorphic limit $p = \mathcal{P}_{\text{roots}}(A_0, B_0)$. Uniqueness in RLA5 then proves

$$(\alpha, \beta)(\mathcal{P}_{\text{roots}}(A_0, B_0)) = (A_0, B_0).$$

The other inverse composition follows from uniqueness in RLA9. This explicitly restores z, x, u, a from the spectral-root coordinates.

For a parameter function write $h^{\sharp}(p) = \overline{h(\bar{p})}$. Since $\overline{\omega(-\bar{t})} = \omega(t)$, the exact reflection and real-slice restrictions are

$$F_-(t, p) = \overline{F_+(-\bar{t}, \bar{p})}, \quad \beta = -\alpha^{\sharp}, \quad U(-\bar{t}, \bar{p}) = \overline{U(t, p)}, \quad \beta(p) = -\overline{\alpha(p)} \quad (p \in \mathbb{R}^2). \tag{RLA10}$$

The first identity and uniqueness in RLA5 prove the second. Reflecting both linear factors in RLA6, including their minus signs, proves the identity for U . On the real slice $U(0, p)$ is real and stays negative, since its value at p_* is $-b$. Accordingly $g_0 = -U(0, p)|\alpha(p)|^2 > 0$ away from p_* . The spectral collision curve $\alpha = \beta$ is a different curve from the parameter branches $\alpha = 0$, $\beta = 0$. At its nonzero points $g_0 \neq 0$, so collision of the two nonzero spectral roots does not make the finite conductor singular.

1.3 Every cutoff and the full two-generator module

Let R be the local ring of convergent holomorphic parameter germs at p_* . RLA9 gives mutually inverse ring identifications $R \simeq \mathbb{C}\{\alpha, \beta\}$; the original parameter maps and unit remain those already displayed. Put $\sigma = \alpha + \beta$, $\pi = \alpha\beta$, $R_N = R[t]/(t^N)$ for $N \geq 1$ and $q = t^2 - \sigma t + \pi$. Write $L_N(h)$ for multiplication by the full series h modulo t^N in the ordered basis $1, t, \dots, t^{N-1}$. Each entry uses the exact coefficient, not a truncated substitute for that coefficient. Since $U(0, p)$ is a unit, the finite inverse of $L_N(U)$ is multiplication by U^{-1} modulo t^N ; its coefficients are specified recursively by

$$v_0 = U_0^{-1}, \quad v_n = -U_0^{-1} \sum_{j=1}^n U_j v_{n-j}, \quad (\text{RLA11})$$

$$L_N(g) = L_N(q)L_N(U), \quad \mathcal{M}_N := \text{coker } L_N(g) = R[t]/(t^N, q).$$

Here $U_j = [t^j]U$, and the equality of cokernels uses the identity on the target: $(g) = (q)$ in R_N , because multiplication by U is invertible there. The corresponding source morphism is $L_N(U)$ and its exact inverse is the displayed recurrence. Its metric effect is calculated below, rather than declared isometric.

Define $h_{-1} = 0$, $h_0 = 1$ and

$$h_n(\alpha, \beta) = \sum_{j=0}^n \alpha^{n-j} \beta^j \quad (n \geq 0), \quad h_n = \sigma h_{n-1} - \pi h_{n-2} \quad (n \geq 1), \quad C = \begin{pmatrix} 0 & -\pi \\ 1 & \sigma \end{pmatrix}. \quad (\text{RLA12})$$

The recurrence for $n = 1$ means $h_1 = \sigma$; for $n \geq 2$ it follows by multiplying and cancelling the finite sums. Polynomial division by the monic q is exact over R and leaves a unique representative $a + bt$. Indeed each power t^n with $n \geq 2$ is reduced using $t^2 = \sigma t - \pi$, and a nonzero multiple of the monic quadratic has degree at least two. Multiplication by t on this rank-two module has the matrix C . Induction gives the complete remainders

$$[t^n]_q = -\pi h_{n-2} + h_{n-1}t \quad (n \geq 1), \quad C^N = \begin{pmatrix} -\pi h_{N-2} & -\pi h_{N-1} \\ h_{N-1} & h_N \end{pmatrix} \quad (N \geq 1). \quad (\text{RLA13})$$

The first formula at $n = 1$ uses $h_{-1} = 0$. The columns of C^N are exactly the remainders of t^N, t^{N+1} . Every multiple of t^N in $R[t]/q$ is their R -linear combination, since that ring has basis $1, t$. Hence the full conductor has the two-generator presentation

$$R^2 \xrightarrow{C^N} R^2 \longrightarrow \mathcal{M}_N \longrightarrow 0, \quad [a + bt] \longmapsto [a + bt] \bmod (t^N, q). \quad (\text{RLA14})$$

The inverse to its cokernel map reduces any polynomial modulo q by RLA13. Reducing a multiple of t^N changes its remainder by C^N times a vector, and reducing a multiple of q changes it by zero. These facts prove well-definedness and both compositions. Conversely, a remainder in the image of C^N is a multiple of t^N modulo q , so the same argument proves injectivity. Thus RLA14 describes the entire module, not only its dimension at the special point. It includes $N = 1$: the unit entry of C leaves $\mathcal{M}_1 = R/(\pi)$.

In the displayed presentation the determinant and entry ideals are

$$\text{Fitt}_0(\mathcal{M}_N) = (\pi^N), \quad \text{Fitt}_1(\mathcal{M}_N) = (h_{N-1}, \pi h_{N-2}), \quad \text{Fitt}_j(\mathcal{M}_N) = R \quad (j \geq 2). \quad (\text{RLA15})$$

Here Fitting ideals mean the ideals of minors of sizes two, one and zero in RLA14. The determinant is $(\det C)^N = \pi^N$; $h_N = \sigma h_{N-1} - \pi h_{N-2}$ gives the entry ideal. For $N = 1$ it is R , since $h_0 = 1$. These are also the corresponding ideals from the original N -generator presentation: adding a generator with a relation that solves uniquely for it, or eliminating such a pair, adjoins or removes an identity block. The minors of that enlarged block give precisely

the same ideals with their shifted sizes. Multiplication of a presentation by invertible matrices leaves its minors' ideals fixed by the determinant expansion, applied also to the inverse matrices. For an explicit common presentation start with generators $1, t, \dots, t^{N+1}$ and relations $q, tq, \dots, t^{N-1}q, t^N, t^{N+1}$. Eliminating the last two generators with their unit-coefficient relations gives the original multiplication-by- q presentation modulo t^N . Instead eliminating $t^2, \dots, t^{N+1}$ in order with the first N monic relations leaves $1, t$ and precisely the two columns of C^N . These are the indicated invertible generator and relation operations; no ideal is inferred from dimensions alone. For $N \geq 2$ the common zero of the two generators of Fitt_1 near the origin is just $\alpha = \beta = 0$: if $\pi = 0$ and one root is nonzero, h_{N-1} is its nonzero $(N-1)$ st power; if $\pi \neq 0$, consecutive h 's cannot both vanish, as the recurrence backwards would force $h_0 = 0$.

1.4 The exact gluing of the two branches

Evaluation at the two roots is an injective map on $R[t]/q$:

$$a + bt \mapsto (a + b\alpha, a + b\beta), \quad \mathcal{L} = \{(v_+, v_-) \in R^2 : v_+ - v_- \in (\alpha - \beta)\}. \quad (\text{RLA16})$$

Its image is exactly $\mathcal{L}$: divide $v_+ - v_-$ by $\alpha - \beta$ to recover b , and then set $a = v_+ - b\alpha$. Its kernel is zero since $(\alpha - \beta)b = 0$ in the integral domain of convergent series. Multiplication by t^N becomes $\text{diag}(\alpha^N, \beta^N)$ on this lattice. It follows, with all maps specified, that

$$\begin{aligned} 0 \longrightarrow \mathcal{M}_N \xrightarrow{\iota} R/(\alpha^N) \oplus R/(\beta^N) \xrightarrow{d} R/(\alpha - \beta, \alpha^N) \longrightarrow 0, \\ \iota([P]) = (P(\alpha) \bmod \alpha^N, P(\beta) \bmod \beta^N), \quad d([v_+], [v_-]) = [v_+ - v_-]. \end{aligned} \quad (\text{RLA17})$$

The last map is well defined and onto. Its kernel consists of pairs whose representatives differ by an element of $(\alpha - \beta, \alpha^N)$; changing the first representative by a multiple of α^N puts them in $\mathcal{L}$, proving that the kernel is the image of ι . For injectivity, suppose $(\alpha^N r, \beta^N s) \in \mathcal{L}$. Modulo $\alpha - \beta$ the relation is $\alpha^N(r - s) = 0$ in $\mathbb{C}\{\alpha\}$. This ring is a domain, so $r - s$ is divisible by $\alpha - \beta$. Thus $(r, s) \in \mathcal{L}$, and the original pair is a multiple of t^N in the lattice, proving that its class is zero. This proves exactness directly, including the intersection coupling. The gluing quotient has complex dimension N , with basis $1, \alpha, \dots, \alpha^{N-1}$ after setting $\beta = \alpha$. The module is not the direct sum of the two branch modules for any $N \geq 1$. For $N = 1$ its minimal generator count is one, whereas that direct sum needs two, as reduction modulo (α, β) shows. For $N \geq 2$, RLA15's first Fitting ideal contains h_{N-1} , a nonzero homogeneous polynomial of degree $N - 1$. The first Fitting ideal of the diagonal presentation of $R/(\alpha^N) \oplus R/(\beta^N)$ is (α^N, β^N) ; every element has order at least N . Those ideals differ, so the presentation invariance just proved excludes an isomorphism. RLA17 retains the exact embedding and gluing map that relates these modules.

The annihilator is exactly

$$\text{Ann}_R(\mathcal{M}_N) = (\alpha^N) \cap (\beta^N) = (\alpha^N \beta^N). \quad (\text{RLA18})$$

Injection in RLA17 gives inclusion from right to left. Each projection of ι is onto, because constant polynomials give every residue in each component; hence an annihilator must lie in both ideals. In the convergent double Taylor series, divisibility by α^N removes all powers of α below N , and divisibility by β^N removes all powers of β below N . Both together give divisibility by their product. At a point of $\alpha = 0$ with $\beta \neq 0$, the second branch module and the gluing quotient in RLA17 vanish locally, so the module is exactly $R/(\alpha^N)$ there. The corresponding statement on the other branch is $R/(\beta^N)$. At their intersection, RLA14 reduced modulo (α, β) has zero matrix for $N \geq 2$, and therefore a two-dimensional fiber. For $N = 1$ the fiber has dimension one, as already calculated. The two dimensions at the intersection record the two original lost directions; the gluing map retains how the branches approach that intersection.

1.5 Exact connection to the directional resonances and the full inverse

Away from $\alpha\beta = 0$ the exact inverse coefficients are

$$[t^n] \frac{1}{g(t, p)} = \sum_{j=0}^n v_j(p) \frac{h_{n-j}(\alpha(p), \beta(p))}{(\alpha(p)\beta(p))^{n-j+1}}, \quad 0 \leq n < N, \quad (\text{RLA19})$$

where every v_j is the full unit reciprocal from RLA11. Indeed the geometric expansions of $(t - \alpha)^{-1}$ and $(t - \beta)^{-1}$ yield $[t^m]q^{-1} = h_m(\alpha, \beta)/(\alpha\beta)^{m+1}$. Finite convolution with U^{-1} proves RLA19. It holds also when $\alpha = \beta \neq 0$, where the polynomial finite sum gives $h_m = (m + 1)\alpha^m$; no divided-difference singularity is inserted. This formula includes every higher unit coefficient and the complete $e^{-4\omega}$ center factor in RLA6.

For a nonzero real vector v in the unchanged (z, x) coordinates, let $A_v = A_z v_z + A_x v_x$ and define the literal derivatives $a_v = -iA_v/B$, $b_v = -i\bar{A}_v/\bar{B} = -\bar{a}_v$. They are nonzero because the real derivative is invertible. Along $p_* + \tau v$, RLA8 gives $\alpha = \tau a_v + O(\tau^2)$ and $\beta = \tau b_v + O(\tau^2)$, while $v_0(p_*) = -1/b$. Consequently RLA19 proves the same exact leading reciprocal coefficient as RPD13–15:

$$H_D(v) = -\frac{h_D(a_v, b_v)}{b(a_v b_v)^{D+1}}, \quad H_D(v) = 0 \iff \frac{a_v}{b_v} = e^{2\pi i k/(D+1)}, \quad 1 \leq k \leq D. \quad (\text{RLA20})$$

For the first equality, the $j = 0$ term in RLA19 has order τ^{-D-2} ; each $j > 0$ term has strictly lower order. For the second, $a_v, b_v \neq 0$ and $(a_v - b_v)h_D(a_v, b_v) = a_v^{D+1} - b_v^{D+1}$. At equality $a_v = b_v$, $h_D = (D + 1)a_v^D \neq 0$; otherwise the displayed roots of unity are all possibilities. Since $a_v/b_v = (A_v/B)/(A_v/B)$, this recovers exactly the original resonance lines of RPD20, including their orientation and original scale. Thus the old reciprocal resonance is the vanishing of the degree- D remainder coefficient of the actual two-root algebra, and the other entry of the presentation remains. RLA19, rather than just its leading term, retains the curvature and unit corrections calculated in RPD23–30.

1.6 Original Gamma metric and an exact residual matrix using every coefficient

For $s \in \{1, a_{\text{amp}}\}$, $c_0 = a_{\text{amp}}/2$, $c_1 = c_0 - 4$, retain the original data

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \quad \rho_n = \sqrt{n!/(s/2)_n}, \\ p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = y p_{n,s} - n(n-1 + s/2) p_{n-1,s}, \\ h_{n,s} &= M_s n! (s/2)_n, \quad \phi_{n,s,c}(S) = p_{n,s}((S - c)/i)/\sqrt{h_{n,s}}, \quad \|P\|_{s,c}^2 = \int |P(c + iy)|^2 dm_s(y). \end{aligned} \quad (\text{RLA21})$$

These are WCF2's exact original orthonormal frames, with no mass or center replacement. For $D = N - 1$ set $D_\rho = \text{diag}(\rho_0, \dots, \rho_D)$ and let J_N reverse these N indices. If x is the source orthonormal coefficient column, the exact polynomial coordinate morphism and inverse are

$$x \mapsto a(t) = \sum_{j=0}^{N-1} (J_N D_\rho x)_j t^j, \quad x = D_\rho^{-1} J_N \text{coeff}(a), \quad B_{D,s} = D_\rho^{-1} J_N L_N(g) J_N D_\rho. \quad (\text{RLA22})$$

The same formula uses the target frame at c_1 on the output. Since $L_N(g)_{ij} = g_{i-j}$ for $i \geq j$, reversal makes its entries g_{j-i} above the diagonal, proving RLA22 entry by entry. The original Gamma norm in these polynomial coordinates is

$$a^* W_{N,s} a, \quad W_{N,s} = \text{diag}(\rho_{N-1}^{-2}, \dots, \rho_0^{-2}). \quad (\text{RLA23})$$

Here a in the quadratic form means its coefficient column; the original unit still has the distinct meaning specified in RLA1. Formula RLA23 is the exact transport of the original normalized frame coefficients, not an assertion that the physical measure has lost its mass. Its inverse in RLA22 uses the full $h_{n,s}$ and centers in RLA21 when it reconstructs the physical polynomial. For the source change $c = L_N(U)a$ in RLA11, the source Gram becomes

$$(L_N(U)^{-1})^* W_{N,s} L_N(U)^{-1}, \quad (\text{RLA24})$$

whereas the target Gram remains $W_{N,s}$. This explicitly accounts for every unit coefficient that affects the cutoff; no multiplication by a nonconstant unit is asserted to be an isometry.

There is also a two-by-two residual presentation directly in the original orthonormal matrix, retaining all the higher g_j . For $N \geq 2$ order the columns as $(2, \dots, N-1, 0, 1)$ and keep the row order $(0, \dots, N-1)$. With the corresponding permutation matrix Π , write

$$B_{D,s}\Pi = \begin{pmatrix} P & Q \\ R_0 & S_0 \end{pmatrix}, \quad P_{ij} = B_{i,j+2} \quad (0 \leq i, j \leq N-3), \quad \Sigma_N = S_0 - R_0 P^{-1} Q. \quad (\text{RLA25})$$

The remaining blocks are the literal remaining rows and columns, so this formula specifies every entry using RLA22. At the base point P is upper triangular with nonzero diagonal $-b\rho_{i+2}/\rho_i$. Thus $\det P$ is a unit germ and $P^{-1} = \text{adj}(P)/\det P$ is an exact finite formula. For $N = 2$ the pivot block is empty and $\Sigma_2 = B_{1,s}$. The two invertible elimination matrices satisfy

$$E_L = \begin{pmatrix} I & 0 \\ -R_0 P^{-1} & I_2 \end{pmatrix}, \quad E_R = \begin{pmatrix} P^{-1} & -P^{-1} Q \\ 0 & I_2 \end{pmatrix}, \quad E_L(B_{D,s}\Pi)E_R = \text{diag}(I_{N-2}, \Sigma_N). \quad (\text{RLA26})$$

Block multiplication proves this identity and its inverse; all entries, including $g_3, \dots, g_{N-1}$, enter P, Q, R_0, S_0 . It proves directly $\text{coker } \Sigma_N \simeq \mathcal{M}_N$. Moreover the column permutation has sign $(-1)^{2(N-2)} = 1$, so

$$\det \Sigma_N = \frac{g_0^N}{\det P} = \frac{U(0,p)^N (\alpha\beta)^N}{\det P}, \quad G_{\text{source}} = E_R^* E_R, \quad G_{\text{target}} = (E_L^{-1})^* E_L^{-1}. \quad (\text{RLA27})$$

Indeed original source coordinates are ΠE_R times the new ones and original target coordinates are E_L^{-1} times the new ones; the original frames are orthonormal. These complete Grams include all cross terms between the pivot coordinates and the two residual coordinates. For example every squared original singular value is exactly a root in λ of

$$\det(\text{diag}(I, \Sigma_N)^* G_{\text{target}} \text{diag}(I, \Sigma_N) - \lambda G_{\text{source}}) = 0. \quad (\text{RLA28})$$

This is the defining eigenvalue equation for the transported Hermitian form and follows by multiplying by an invertible square root of G_{source} . Thus the two-generator algebra and the original analytic norm are connected by exact finite morphisms; the residual matrix alone is never substituted for the full metric. At $N = 1$ the original matrix is the scalar g_0 , already covered by RLA7 and RLA11–18.

1.7 All three conductor domains and the four original ES labels

The exact order $v = \text{ord}_{t=0} g(t, p)$ is determined on the whole local complex parameter space by

parameter locus	v	g_v
$\alpha\beta \neq 0$	0	$U_0\alpha\beta$
$\alpha = 0, \beta \neq 0$	1	$-U_0\beta$
$\beta = 0, \alpha \neq 0$	1	$-U_0\alpha$
$\alpha = \beta = 0$	2	$U_0 = -b$

(RLA29)

All entries follow by multiplying the complete factorization RLA6. Since $\omega'(0) = -i \neq 0$, this is also the exact order of $E_A(\xi, p)$ at $\xi = 0$, with $g_v = (-i)^v \mu_v / v!$. For every $D \geq 0$ the actual order-adjusted map is therefore

$$\mathcal{A}_{D,s,p} : \mathcal{P}_{D+v}/\mathcal{P}_{v-1} \longrightarrow \mathcal{P}_D, \quad (\mathcal{A}_{D,s,p})_{ij} = \begin{cases} g_{j-i+v}(p) \rho_{j+v}/\rho_i, & j \geq i, \\ 0, & j < i. \end{cases} \quad (\text{RLA30})$$

Its diagonal $g_v \rho_{j+v}/\rho_j$ never vanishes on its stated stratum. The source quotient uses the original norm at c_0 and the target uses the original norm at c_1 . WCF3's generating identity, or its coefficient comparison in WCF4–6, proves this matrix; inversion is the exact weighted triangular reciprocal of $t^{-v}g$. The fixed old map on $\mathcal{P}_D$, in contrast, has kernel $\mathcal{P}_{\min(v,D+1)-1}$ and image $\mathcal{P}_{D-v}$ when $D \geq v$, and zero image when $D < v$. This follows immediately from $T_A S^n = \sum_j \binom{n}{j} \mu_j S^{n-j}$ and nonzero leading $\binom{n}{v} \mu_v$ for $n \geq v$. The two parameter branches have $v = 1$ individually, their intersection has $v = 2$, and the real slice has only $v = 0, 2$ locally. Both changed domains and the old kernel remain explicit.

Retain the four original ES columns K , ordered M, N, T, E, and the exact Lagrange polynomials $\ell_a(S) = \prod_{b \neq a} (S - \rho_b^{\text{prim}}) / (\rho_a^{\text{prim}} - \rho_b^{\text{prim}})$. Let $\Phi_0 q = \sum_a (K^{-1} q)_a \ell_a(S)$; thus $\Phi_0^{-1} P = K(P(\rho_a^{\text{prim}}))_a$. On the stratum of order $v = 0, 1, 2$, the exact augmented old map is

$$\Theta_v q = (T_A \Phi_0 q, ([S^r] \Phi_0 q)_{0 \leq r < v}) \in \mathcal{P}_{3-v} \oplus \mathbb{C}^v. \quad (\text{RLA31})$$

It is an isomorphism. To give its inverse for every stratum, write the supplied target as $Y = \sum_{k=0}^{3-v} Y_k S^k$ and set $c_r = t_r$ for $0 \leq r < v$. Solve successively, in descending order $k = 3-v, \dots, 0$, by the exact finite equations

$$c_{k+v} = \frac{Y_k - \sum_{n=k+v+1}^3 \binom{n}{k} \mu_{n-k} c_n}{\binom{k+v}{k} \mu_v}, \quad \Theta_v^{-1}(Y, t) = K \left(\sum_{n=0}^3 c_n (\rho_a^{\text{prim}})^n \right)_a. \quad (\text{RLA32})$$

All denominators are nonzero by RLA29. The coefficient of S^k in $T_A \sum c_n S^n$ is the numerator equation used to solve for c_{k+v} ; the lower coefficients $c_0, \dots, c_{v-1}$ contribute zero. This proves the two inverse compositions by descending substitution and interpolation. At $v = 2$, it is exactly the RPE20–21 map, with the two original lost scalars t_0, t_1 . At $v = 1$ it reads explicitly

$$c_3 = \frac{Y_2}{3\mu_1}, \quad c_2 = \frac{Y_1 - 3\mu_2 c_3}{2\mu_1}, \quad c_1 = \frac{Y_0 - \mu_2 c_2 - \mu_3 c_3}{\mu_1}, \quad c_0 = t_0. \quad (\text{RLA33})$$

Thus each individual branch loses one scalar and the intersection loses the two-dimensional old kernel, without merging any ES label. For every v the four complete states are

$$\Theta_v(K e_a) = (T_A \ell_a, ([S^r] \ell_a)_{r < v}), \quad x_a^{(v)} = [S^v \ell_a] \in \mathcal{P}_{v+3}/\mathcal{P}_{v-1}, \quad y_a^{(v)} = T_A(S^v \ell_a). \quad (\text{RLA34})$$

The four $x_a^{(v)}$ are independent because division by S^v on the representative with degrees v through $v+3$ recovers the independent Lagrange basis. RLA30 then proves independence of the four corrected outputs. With $X_v e_a = x_a^{(v)}$, $Y_v e_a = y_a^{(v)}$, the exact morphisms remain $T_A X_v K^{-1} = Y_v K^{-1}$, and conjugation of the unchanged ES polynomial through these invertible maps gives the same original intertwining as RPE18. RLA31–33 separately recover the old fixed source with its lost scalars.

For completeness every original metric cross term in this recovery is fixed by a finite formula. Let S_v be the coefficient matrix of RLA32 from $(Y_0, \dots, Y_{3-v}, t_0, \dots, t_{v-1})$ to (c_0, c_1, c_2, c_3) . Put $m_0 = 1$, $m_2 = s/2$, $m_4 = (s/2)(3s/2 + 2)$, $m_6 = (s/2)(15(s/2)^2 + 30(s/2) + 16)$, and $m_1 = m_3 = m_5 = 0$. Then the complete transported source Gram is

$$G_{\Theta_v} = S_v^* H_{s,c_0} S_v, \quad (H_{s,c})_{jk} = M_s \sum_{a=0}^j \sum_{d=0}^k \binom{j}{a} \binom{k}{d} c^{j+k-a-d} (-i)^a i^d m_{a+d}, \quad 0 \leq j, k \leq 3. \quad (\text{RLA35})$$

The finite sum expands $\int \overline{(c + iy)^j} (c + iy)^k dm_s(y)$. The six moments are those obtained from WCF2's original recurrence and norms, as calculated in RPE22–23; substituting them gives every entry, including the source mass and all cross terms. The output metric uses the corresponding upper-left $(4 - v) \times (4 - v)$ block of H_{s, c_1} instead. Thus neither the original target metric nor either lost scalar is silently replaced by a quotient coordinate metric.

Equations RLA2–35 construct one complete local object: the actual original analytic family with its invertible parameter maps, its two transverse parameter branches, its full analytic spectral units, and every finite weighted conductor and branch-glued cokernel. All roots, branches and domains belong to this certified geometric member. No equation identifies it with an arithmetic zeta quartet.

2 The moving kernel, its exact metric, and the original inverse bound

The local two-branch construction RLA1–35 now permits a single finite-dimensional comparison across its three actual order strata. The complete original operator and orthonormal frames are WCF1–6 [10]; the certified point and four original labels remain RPZ1–19 and RPE1–26 [9, 8]. All directional exponents used below are the proved RPD23–30 [6]. We derive each new matrix and metric identity. The construction contains an explicitly stated orthogonal projection; it does not identify a projected extension with the original operator.

Fix an integer $D \geq 1$, put $N = D + 1$, and keep $s \in \{1, a_{\text{amp}}\}$. All neighborhoods below may depend on these fixed choices. They are not assertions uniform in the amplification or cutoff. Keep ρ_j , the full mass, centers and physical frames of RLA21. Let P_D denote orthogonal projection onto the first $D + 1$ target frames, and define

$$\mathcal{R}_p = P_D T_A|_{\mathcal{P}_{D+2}}, \quad R(p)_{ij} = \begin{cases} g_{j-i}(p) \rho_j / \rho_i, & j \geq i, \\ 0, & j < i, \end{cases} \quad 0 \leq i \leq D, \quad 0 \leq j \leq D+2. \quad (\text{RQT1})$$

This entry formula follows by coefficient comparison in WCF3, with the unchanged factor $g = e^{-4\omega} E_A$ from RLA2. Projection discards exactly target degrees $D + 1, D + 2$; every coefficient of the retained entries is the original coefficient. On the actual order- v source $\mathcal{P}_{D+v}$, $0 \leq v \leq 2$, the original operator already has degree at most D , so that projection acts as the identity on its output.

Partition the source frames after degrees zero and one:

$$R = [L \ H], \quad L \in \mathbb{C}^{N \times 2}, \quad H \in \mathbb{C}^{N \times N}, \quad H_{ij} = g_{j+2-i} \rho_{j+2} / \rho_i \quad (0 \leq i, j < N), \quad (\text{RQT2})$$

where $g_k = 0$ for $k < 0$. At p_* the coefficients g_0, g_1 vanish and $g_2 = -b \neq 0$. Thus $L(p_*) = 0$ and $H(p_*)$ is upper triangular with diagonal $-b \rho_{i+2} / \rho_i$. It is exactly the original order-two quotient matrix, including all $g_3, \dots, g_{D+2}$. Its nonzero determinant gives a neighborhood on which H is invertible. Define the exact matrices

$$\begin{aligned} Z &= H^{-1} L, & M &= (I_N + Z Z^*)^{-1}, & W &= (I_2 + Z^* Z)^{-1}, \\ K &= \ker R = \left\{ \begin{pmatrix} l \\ -Zl \end{pmatrix} : l \in \mathbb{C}^2 \right\}. \end{aligned} \quad (\text{RQT3})$$

The last equality follows from $R(l, h) = H(Zl + h)$, and proves surjectivity as well as the exact kernel. In particular $K(p_*)$ is the original $\mathcal{P}_1$ inside $\mathcal{P}_{D+2}$. Off the order-two locus it is the kernel of the projected extension; it is not called the kernel of the full T_A .

2.1 The quotient and both exact norm transports

The map from a source vector to its quotient coordinate is $w = h + Zl$. Vectors with that same w differ by K . Minimizing the original squared norm $\|l\|^2 + \|w - Zl\|^2$ gives

$$l = Z^* M w, \quad h = M w, \quad \|[l, h]\|_{\mathcal{P}_{D+2}/K}^2 = w^* M w, \quad R^\dagger = \begin{pmatrix} Z^* \\ I_N \end{pmatrix} M H^{-1}. \quad (\text{RQT4})$$

For a direct proof, the minimum solves $(I_2 + Z^* Z)l = Z^* w$. The identity $(I_2 + Z^* Z)Z^* = Z^*(I_N + Z Z^*)$ gives the displayed l ; then $h = w - Zl = M w$. Subtracting this pair from any other representative gives $(u, -Zu)$, which is orthogonal to the minimizer. Its squared norm is nonnegative and vanishes only at zero. This proves the minimum, its uniqueness, and the norm. It also proves $RR^\dagger = I_N$ and that $R^\dagger y$ is the unique preimage of y orthogonal to K .

Let square roots be the positive Hermitian roots, and set

$$Q_+ = \begin{pmatrix} Z^* \\ I_N \end{pmatrix} M^{1/2}, \quad Q_K = \begin{pmatrix} I_2 \\ -Z \end{pmatrix} W^{1/2}, \quad \tilde{A} = R Q_+ = H M^{-1/2}, \quad \tilde{A}^{-1} = M^{1/2} H^{-1}. \quad (\text{RQT5})$$

Multiplication gives $Q_+^* Q_+ = I_N$, $Q_K^* Q_K = I_2$ and $Q_K^* Q_+ = 0$. Their columns consequently form an orthonormal basis of the entire original source space. In particular $Q_+ Q_+^* + Q_K Q_K^* = I_{N+2}$. This proves that Q_+ realizes the quotient isometrically, with precisely the metric RQT4. It is a proved coordinate isometry between these two stated spaces, not a rescaling of the original measure or operator.

Write $m(p)$ and $M_R(p)$ for the smallest and largest singular values of R (there are exactly N positive ones). Then

$$RR^* = H(I_N + Z Z^*)H^* = \tilde{A} \tilde{A}^*, \quad m \geq \|H^{-1}\|^{-1}, \quad M_R \leq \|H\| \sqrt{1 + \|Z\|^2}. \quad (\text{RQT6})$$

Indeed the middle positive matrix is at least I_N , and its largest eigenvalue is $1 + \|Z\|^2$; apply the squared norm identity to each vector. Shrinking the neighborhood so that $\|H - H(p_*)\| \leq (2\|H(p_*)^{-1}\|)^{-1}$ gives $m \geq (2\|H(p_*)^{-1}\|)^{-1}$. This follows by expanding $(I + H(p_*)^{-1}(H - H(p_*)))^{-1}$ as its convergent geometric matrix series. Thus the moving quotient inverse is bounded on that neighborhood, with the original finite matrix constant.

2.2 Every actual-order domain maps into that quotient

Let $J_v : \mathbb{C}^N \rightarrow \mathbb{C}^{N+2}$ insert a vector in source positions $v, \dots, D + v$, for $v = 0, 1, 2$. On the stratum of actual order v , these are the original orthonormal quotient representatives for $\mathcal{P}_{D+v}/\mathcal{P}_{v-1}$, so define

$$A_v = R J_v, \quad T_v = Q_+^* J_v, \quad E_v = T_v^* T_v = J_v^* (I - Q_K Q_K^*) J_v, \quad A_v = \tilde{A} T_v. \quad (\text{RQT7})$$

These are exact morphisms, not just determinant identities. The proof uses $R Q_K = 0$ and the complete orthogonal decomposition following RQT5. On its stated stratum, A_v is the actual RLA30 matrix, with diagonal $g_v \rho_{j+v}/\rho_j$. Consequently T_v is invertible there. The map induced by orthogonal projection from the actual-order source into $\mathcal{P}_{D+2}/K$ has squared source metric E_v . The original source metric itself remains I_N .

Its exact determinant is

$$\det E_v = \frac{|g_v|^{2N}}{\det(RR^*)} \prod_{j=0}^D \frac{\rho_{j+v}^2}{\rho_j^2}, \quad g_v = \frac{(-i)^v \mu_v}{v!}. \quad (\text{RQT8})$$

Take squared absolute determinants of RQT7 and use RQT6. This also proves that every norm correction between the actual quotient and the moving quotient is retained, not suppressed. At p_* , $Z = 0$ and $J_2 = Q_+$; hence $T_2 = I_N$, $E_2 = I_N$, and $\tilde{A} = A_2 = H$. At order zero or one, RQT7–8 give the full correction instead of asserting those metrics equal.

2.3 Exactly two angles control the old source

Let J_{tail} insert in the last two positions $D+1, D+2$, the orthogonal complement of the old source $J_0 \mathbb{C}^N$. Split Z into its first $N-2$ and last two rows:

$$Z = \begin{pmatrix} Z_{\text{top}} \\ Z_{\text{bot}} \end{pmatrix}, \quad V = J_{\text{tail}}^* Q_K = -Z_{\text{bot}} W^{1/2}. \quad (\text{RQT9})$$

The singular values of T_0 are exactly

$$\underbrace{1, \dots, 1}_{N-2}, \quad d_1, d_2, \quad 1 \geq d_1 \geq d_2 \geq 0, \quad (d_1, d_2) = \text{singular values}(V). \quad (\text{RQT10})$$

To prove it, put $F = J_0^* Q_K$. Then $E_0 = I_N - FF^*$. The nonzero eigenvalues of FF^* and F^*F agree, with multiplicity: the maps F and F^* intertwine their eigenspaces at each nonzero eigenvalue and are inverse there up to that eigenvalue. Meanwhile $F^*F = Q_K^*(I - J_{\text{tail}} J_{\text{tail}}^*) Q_K = I_2 - V^*V$. The remaining $N-2$ eigenvalues of E_0 are one. This proves RQT10 including endpoints and multiplicities, also when an extra one or zero occurs. Geometrically d_j are the sines of the two angles between K and the old source: the formula above is their exact definition through orthogonal projections in the original Gamma metric. Both are zero at p_* .

There is an exact two-by-two inverse calculation with all higher coefficients present. A vector in the old source has full source coordinates $(l, m, 0_2)$, so

$$H^{-1} B_{D,s}(l, m) = \begin{pmatrix} Z_{\text{top}} l + m \\ Z_{\text{bot}} l \end{pmatrix}, \quad \det Z_{\text{bot}} = \frac{g_0^N}{\det H}. \quad (\text{RQT11})$$

Here the determinant is valid also at singular parameters. For proof, permute the old columns from (l, m) to (m, l) ; the sign is $(-1)^{2(N-2)} = 1$. The resulting matrix is $\begin{pmatrix} I & Z_{\text{top}} \\ 0 & Z_{\text{bot}} \end{pmatrix}$, and $\det B_{D,s} = g_0^N$. At $g_0 \neq 0$, if $H^{-1}y = (w_{\text{top}}, w_{\text{bot}})$, its exact inverse is

$$l = Z_{\text{bot}}^{-1} w_{\text{bot}}, \quad m = w_{\text{top}} - Z_{\text{top}} Z_{\text{bot}}^{-1} w_{\text{bot}}. \quad (\text{RQT12})$$

Substitution proves both inverse compositions. In particular every divergence is accounted for by this explicit two-by-two matrix and the bounded matrices around it; no coefficient has been dropped. The exact product of the two angle sines is

$$d_1 d_2 = \frac{|g_0|^N}{\sqrt{\det(RR^*)}} = \frac{|U(0, p)|^N |\alpha \beta|^N}{|\det H| \sqrt{\det(I_2 + Z^* Z)}}. \quad (\text{RQT13})$$

This follows from RQT8 with $v = 0$ and RQT10. The second form uses RLA7 and RQT6; the equality $\det(I_N + ZZ^*) = \det(I_2 + Z^* Z)$ follows from the same nonzero-eigenvalue correspondence used in RQT10.

For $1 \leq r \leq N$ put $\mathcal{D}_1 = d_2^{-1}$ and $\mathcal{D}_r = (d_1 d_2)^{-1}$ for $r \geq 2$. Off the branches, the exact factorization and bounds are

$$B_{D,s}^{-1} = T_0^{-1} \tilde{A}^{-1}, \quad \frac{\mathcal{D}_r}{\|\bigwedge^r \tilde{A}\|} \leq \left\| \bigwedge^r B_{D,s}^{-1} \right\| \leq \mathcal{D}_r \left\| \bigwedge^r \tilde{A}^{-1} \right\|. \quad (\text{RQT14})$$

The r th exterior norm of T_0^{-1} is $\mathcal{D}_r$ by RQT10, since exterior singular values are the products of the corresponding ordinary singular values. Submultiplicativity applied to RQT7 gives the upper bound; applying it to $T_0^{-1} = B_{D,s}^{-1} \tilde{A}$ gives the lower one. All multipliers here are finite original matrix norms. They may be bounded further by m^{-r} and M_R^r via RQT6.

2.4 The literal relation to inverse-exterior equation (11)

The inverse-exterior estimate under investigation follows from the determinant and forward estimate WCF1–6 [10]. We restate its exact form here, retaining its actual stratum. Put $b_s = s/2$, $n_v = D + v$ and

$$\begin{aligned} A_{\text{abs}}(p) &= \sum_{u,w=0}^8 |a_{uw}(p)|, & R_{n_v,s} &= \begin{cases} (4n_v + 1)^{1/4}, & s = 1, \\ 1, & s = a_{\text{amp}}, \end{cases} \\ U_{D,s}^{(v)} &= e^{1+2\pi\gamma} A_{\text{abs}}(p) R_{n_v,s} (D+1)(2n_v+1)^{4\delta}. \end{aligned} \quad (\text{RQT15})$$

For completeness the forward estimate is obtained on $|t| = r < 1$ by bounding each original power $((1-it)/(1+it))^{(u-4)\delta+i(w-4)\gamma}$ by $e^{2\pi\gamma}((1+r)/(1-r))^{4\delta}$. The fraction has positive real part, modulus between $(1-r)/(1+r)$ and its reciprocal, and argument of absolute value less than $\pi/2$. Cauchy's coefficient formula and $r = m/(m+1)$ give $|g_m| \leq e^{1+2\pi\gamma} A_{\text{abs}}(2m+1)^{4\delta}$ for $m \geq 1$; the constant coefficient obeys the same bound. Use $m = j + v$ in the actual order matrix. Its weight ratios are at most $R_{n_v,s}$: for $s \geq 2$ the sequence ρ_j decreases; for $s = 1$ it increases and $\rho_n \leq (4n+1)^{1/4}$ follows by induction from $(4n+1)(n+1)^2 \leq (4n+5)(n+1/2)^2$. Row and column sums are therefore at most $U_{D,s}^{(v)}$. The weighted Cauchy–Schwarz inequality $|\sum_j a_{ij} x_j|^2 \leq (\sum_j |a_{ij}|) \sum_j |a_{ij}| |x_j|^2$, summed in i , proves $\|A_v\| \leq U_{D,s}^{(v)}$.

Multiplying the singular values and the triangular diagonal gives the requested bound, and RQT8 rewrites it exactly as

$$\begin{aligned} \left\| \bigwedge^r A_v^{-1} \right\| &= \frac{\left\| \bigwedge^{N-r} A_v \right\|}{|\det A_v|} \\ &\leq (U_{D,s}^{(v)})^{N-r} \left| \frac{\mu_v}{v!} \right|^{-N} \prod_{j=0}^D \prod_{k=1}^v \left(\frac{j + b_s + k - 1}{j + k} \right)^{1/2} \\ &= \frac{(U_{D,s}^{(v)})^{N-r}}{\sqrt{\det(RR^*)} \sqrt{\det E_v}}. \end{aligned} \quad (\text{RQT16})$$

For the first equality, if the forward singular values are $\sigma_1 \geq \dots \geq \sigma_N > 0$, both sides are $(\sigma_{N-r+1} \dots \sigma_N)^{-1}$. For the last equality, take the positive square root of RQT8. Thus the small divisor in equation (11) is exactly the product of the moving-quotient volume and its section-volume correction. On the old order-zero source, $\sqrt{\det E_0} = d_1 d_2$. At the certified order-two point, $E_2 = I$ and $\mu_2/2 = b > 0$, so equation (11) is finite for its actual domain. There is no substitution of a zero μ_0 into an order-two formula. Equally, the bounded moving quotient does not bound the old inverse without its singular section correction.

On any real nonzero direction v in the original (z, x) coordinates, $g_0(p_* + \tau v) = c(v)\tau^2 + O(\tau^3)$ with $c(v) = |A_z v_z + A_x v_x|^2 > 0$. Therefore RQT13 proves

$$\lim_{\tau \rightarrow 0} |\tau|^{-2N} d_1 d_2 = \frac{c(v)^N}{b^N} \prod_{j=0}^D \frac{\rho_j}{\rho_{j+2}}. \quad (\text{RQT17})$$

The constant is the full inverse determinant of $H(p_*)$ times $c(v)^N$; its weights have not been omitted. RQT14 and RPD23–30 also give the separate angle exponents: away from a resonant direction they are $(D, D+2)$; on a resonant straight direction they are $(D+1, D+1)$; on the previously constructed turning paths with $0 < q < 1$ they are $(D+q, D+2-q)$, in descending angle order. To justify transfer of each exponent, multiplication by the invertible $\tilde{A}$ and its bounded inverse bounds each singular value by fixed positive factors. This follows from the minimum–maximum characterization of singular values applied to the norm inequalities $m\|T_0 x\| \leq \|B_{D,s} x\| \leq M_R\|T_0 x\|$. RQT17 fixes their product constant; individual constants are computed by the exact matrices RQT9–12, not asserted unchanged by multiplication by $\tilde{A}$.

2.5 The four ES states and the scalar endpoint

For $D = 3$ let X_0 be the matrix of the four Lagrange polynomials ℓ_a in the original source frames, ordered M, N, T, E. Their images in the single moving quotient, and their original outputs, are exactly

$$\widehat{X}_0 = T_0 X_0, \quad Y_0 = B_{3,s} X_0 = \widetilde{A} \widehat{X}_0. \quad (\text{RQT18})$$

No formal label changes. Since X_0 is invertible, the only rank loss here is the computed rank loss of T_0 , which at p_* is two. RLA31–35 recovers the two lost scalar coordinates and their complete metric. On each actual stratum, the corrected inputs $[S^v \ell_a]$ have their own invertible frame matrix X_v ; exactly the same construction gives $Y_v = \widetilde{A} T_v X_v$. This proves the exact bridge from all four ES states to RQT16, while preserving both the fixed old inputs and the original order-adjusted inputs.

The case $D = 0$ needs no artificial two-angle interpretation. Here $R = (g_0, g_1 \rho_1 / \rho_0, g_2 \rho_2 / \rho_0)$ is one row, its moving quotient has dimension one and

$$d_0 = \frac{|g_0|}{\|R\|}, \quad |B_{0,s}^{-1}| = \frac{1}{|g_0|}, \quad \det E_v = \frac{|g_v|^2 \rho_v^2}{\rho_0^2 \|R\|^2} \quad (v \text{ the actual order}). \quad (\text{RQT19})$$

These are obtained by projecting each one-dimensional coordinate section onto the span of R^* . At p_* , R has only its last entry nonzero and $E_2 = 1$. Thus the scalar endpoint and every higher cutoff are included without changing their dimensions.

2.6 The exact constants of both disappearing angles

This calculation strengthens RQT17 from exponent comparison to the complete leading constants. It uses the original weighted matrices RQT1–14 and the complete directional inverse expansions RPD13–30 [6]; the actual root is RPZ1–19 [9]. The original Gamma frames and centers remain WCF1–6 [10], with their pinned complete proof. No bounded matrix is treated as preserving an individual singular-value constant. Its limiting action is multiplied out explicitly.

Fix $D \geq 1$, $N = D + 1$, and $s \in \{1, a_{\text{amp}}\}$, with all original $\rho_j = \sqrt{j!/(s/2)_j}$. Write $H_* = H(p_*)$ for the full order-two matrix in RQT2, and retain $e = g_2(p_*) = -b$, where $b = |G_\xi(0, p_*)|^2 > 0$, and $f = g_3(p_*)$ with its complete center correction. Thus its entries are $(H_*)_{ij} = g_{j+2-i}(p_*) \rho_{j+2} / \rho_i$ for $j \geq i$, and zero for $j < i$. Let $\mathbf{v} = (v_z, v_x) \neq 0$ be a real direction, distinct from the integer order of a stratum, and put $c_{\mathbf{v}} = |G_z v_z + G_x v_x|^2 > 0$. All $H_n(\mathbf{v}), K_n(\mathbf{v})$ below are the exact directional coefficients of RPD13, including all original derivatives.

Define the positive product constant

$$\mathfrak{J}_{D,s}(\mathbf{v}) = \frac{c_{\mathbf{v}}^{D+1}}{b^{D+1}} \frac{\rho_0 \rho_1}{\rho_{D+1} \rho_{D+2}}.$$

This is RQT17's constant: the product of all weights there telescopes because $\prod_{j=0}^D \rho_j / \rho_{j+2} = \rho_0 \rho_1 / (\rho_{D+1} \rho_{D+2})$. The equality also holds for $D = 1$. Along $p = p_* + \tau \mathbf{v}$, if $H_D(\mathbf{v}) \neq 0$, set $\mathfrak{a}_{D,s}(\mathbf{v}) = b |H_D(\mathbf{v})| \rho_{D+2} / \rho_0$. Then the two angle sines in RQT10, ordered $d_1 \geq d_2$, satisfy the exact limits

$$\begin{aligned} \lim_{\tau \rightarrow 0} |\tau|^{-D-2} d_2(p_* + \tau \mathbf{v}) &= \mathfrak{a}_{D,s}(\mathbf{v})^{-1}, \\ \lim_{\tau \rightarrow 0} |\tau|^{-D} d_1(p_* + \tau \mathbf{v}) &= \mathfrak{J}_{D,s}(\mathbf{v}) \mathfrak{a}_{D,s}(\mathbf{v}). \end{aligned} \quad (\text{RQT20})$$

Here τ is real. To prove the constants, RPD14 and RPD18 give the complete matrix limit $\tau^{D+2} B_{D,s}^{-1} \rightarrow H_D(\mathbf{v}) (\rho_D / \rho_0) e_0 e_D^T$. Since $Z(p_*) = 0$, RQT5 gives $\widetilde{A}(p) \rightarrow H_*$. Use the exact identity $T_0^{-1} = B_{D,s}^{-1} \widetilde{A}$. The last row of H_* has only its last entry nonzero, namely $e \rho_{D+2} / \rho_D$. Consequently

$$\tau^{D+2} T_0^{-1} \longrightarrow e H_D(\mathbf{v}) \frac{\rho_{D+2}}{\rho_0} e_0 e_D^T.$$

Taking its norm and using $\|T_0^{-1}\| = d_2^{-1}$ proves the first limit in RQT20. RQT13 gives $d_1 d_2 \sim \mathfrak{J}_{D,s}(\mathbf{v})|\tau|^{2D+2}$; division proves the second. Every factor from the limiting matrix is retained in these constants.

Now let $H_D(\mathbf{v}) = 0$ and write $h = H_{D-1}(\mathbf{v}) \neq 0$, $k = K_D(\mathbf{v})$. For any complex number k_0 define the full matrix

$$\Omega_{D,s}(\mathbf{v}; k_0) = \begin{pmatrix} eh\rho_{D+1}/\rho_0 & (fh + ek_0)\rho_{D+2}/\rho_0 \\ 0 & eh\rho_{D+2}/\rho_1 \end{pmatrix}.$$

It is invertible. If $\sigma_+(\Omega) \geq \sigma_-(\Omega) > 0$ are its singular values, the complete resonant limits are

$$\begin{aligned} \lim_{\tau \rightarrow 0} |\tau|^{-D-1} d_2(p_* + \tau \mathbf{v}) &= \sigma_+(\Omega_{D,s}(\mathbf{v}; k))^{-1}, \\ \lim_{\tau \rightarrow 0} |\tau|^{-D-1} d_1(p_* + \tau \mathbf{v}) &= \sigma_-(\Omega_{D,s}(\mathbf{v}; k))^{-1}, \\ |\det \Omega_{D,s}(\mathbf{v}; k_0)| &= \mathfrak{J}_{D,s}(\mathbf{v})^{-1}. \end{aligned} \tag{RQT21}$$

For a fully scalar evaluation, if Ω has entries $(a_{11}, a_{12}; 0, a_{22})$, put $S_\Omega = |a_{11}|^2 + |a_{12}|^2 + |a_{22}|^2$. Then $\sigma_\pm(\Omega)^2 = (S_\Omega \pm \sqrt{S_\Omega^2 - 4|a_{11}a_{22}|^2})/2$, by the trace and determinant of $\Omega^* \Omega$.

To prove RQT21, the leading corner of $\tau^{D+1} B_{D,s}^{-1}$ from RPD19 is

$$\begin{pmatrix} h\rho_{D-1}/\rho_0 & k\rho_D/\rho_0 \\ 0 & h\rho_D/\rho_1 \end{pmatrix}$$

in rows $0, 1$ and columns $D-1, D$. Its multiplication by H_* uses exactly the last two rows, whose only nonzero submatrix is

$$\begin{pmatrix} e\rho_{D+1}/\rho_{D-1} & f\rho_{D+2}/\rho_{D-1} \\ 0 & e\rho_{D+2}/\rho_D \end{pmatrix}.$$

Their product is exactly $\Omega_{D,s}(\mathbf{v}; k)$ in the same corner. All other entries tend to zero. This proof includes $D = 1$, when both are the complete matrices. The two nonzero singular values of the limit give the limits of $|\tau|^{D+1}/d_2$ and $|\tau|^{D+1}/d_1$, respectively. The RPD15 identity on resonance gives $|h|^2 = b^{D-1}/c_{\mathbf{v}}^{D+1}$. Multiplying the two diagonal entries of Ω therefore proves its determinant formula, independently of k_0 . It agrees exactly with the product constant from RQT13. In particular the term $fh + ek$ in its upper-right entry cannot be discarded when calculating either individual constant.

The same multiplication determines the whole turning transition. Keep the resonant $\mathbf{v}$ and a real transverse direction $\mathbf{w}$, and put $\kappa_D = DH_D(\mathbf{v})[\mathbf{w}] \neq 0$, as proved in RPD24. For $\tau > 0$ use the literal real path $p(\tau) = p_* + \tau(\mathbf{v} + \epsilon(\tau)\mathbf{w})$, with $\epsilon(\tau) \rightarrow 0$. If $\epsilon(\tau)/\tau \rightarrow L \in \mathbb{R}$, the RQT21 limits hold with k replaced by $k + L\kappa_D$. If $\epsilon(\tau) = q_0\tau^q$, $q_0 \neq 0$ and $0 < q < 1$, set $\mathbf{a}_q = b|q_0\kappa_D|\rho_{D+2}/\rho_0$. The complete alternatives are

path	$d_2(p(\tau))$	$d_1(p(\tau))$
$0 < q < 1$	$\mathbf{a}_q^{-1} \tau^{D+2-q} (1 + o(1))$	$\mathfrak{J}_{D,s}(\mathbf{v}) \mathbf{a}_q \tau^{D+q} (1 + o(1))$
$q = 1$	$\sigma_+(\Omega_{D,s}(\mathbf{v}; k + q_0\kappa_D))^{-1} \tau^{D+1} (1 + o(1))$	$\sigma_-(\Omega_{D,s}(\mathbf{v}; k + q_0\kappa_D))^{-1} \tau^{D+1} (1 + o(1))$
$q > 1$	$\sigma_+(\Omega_{D,s}(\mathbf{v}; k))^{-1} \tau^{D+1} (1 + o(1))$	$\sigma_-(\Omega_{D,s}(\mathbf{v}; k))^{-1} \tau^{D+1} (1 + o(1))$.

(RQT22)

The proof uses the uniform matrix formula RPD25, not a fixed-direction limit applied outside its range. When $\epsilon/\tau \rightarrow L$, that formula tends at scale τ^{D+1} to the complete RPD26 corner. Its product with H_* is the displayed Ω with $k + L\kappa_D$, exactly as in RQT21. When $0 < q < 1$, scaling RPD25 instead by τ^{D+2-q} leaves only the top-right coefficient $q_0\kappa_D\rho_D/\rho_0$. Multiplication by the last row of H_* gives coefficient $eq_0\kappa_D\rho_{D+2}/\rho_0$, proving the d_2 limit. The product asymptotic in RQT13 still has constant $\mathfrak{J}_{D,s}(\mathbf{v})$ because the direction tends to $\mathbf{v}$. Division gives the d_1

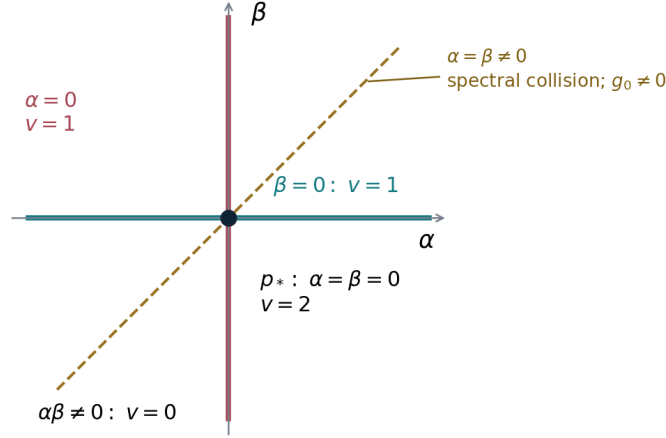
limit. For $q > 1$ the finite-ratio case has $L = 0$. Every limiting corner remains invertible even if $k + L\kappa_D$ or its full combination $fh + e(k + L\kappa_D)$ vanishes. Thus none of these cancellations removes a singular direction. The remaining $N - 2$ singular values of T_0 are exactly one by RQT10.

All constants above belong to the original Gamma metric; in particular ρ_{D+1}, ρ_{D+2} come from the projected extension and are essential. For an original straight (u, x) path the direction and curvature must first be converted by RPD29–30, and its full corrected K_D used in Ω . These equations do not replace that exact coordinate map by a single period-scale factor on a mixed path. As in RLA and RQT, the result concerns this certified geometric member and makes no arithmetic zeta-quartet identification.

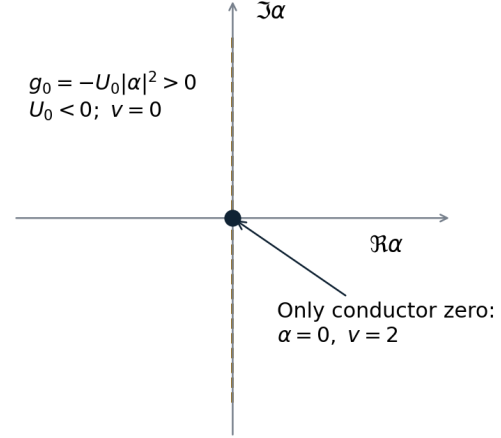
Two branches, one coupled module, and the exact source-angle defect

Schematic local coordinate germs, not a metric embedding or a numerical neighborhood. All original coordinates return through RLA9.

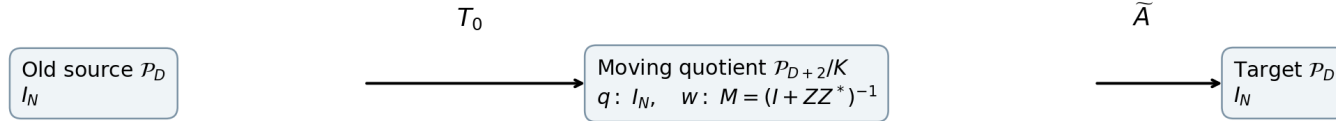
Complexified family: real root-coordinate slice



Actual real parameters: $\beta = -\bar{\alpha}$



The exact comparison in the original Gamma metric (RQT3-16)



$$B = \tilde{A}T_0 \quad \text{sv}(T_0) = (1, \dots, 1, d_1, d_2)$$

$$d_1 d_2 = |g_0|^{D+1} / \sqrt{\det(RR^*)}$$

Proof: RLA6-18, RQT1-19. Original family and Gamma frames: PCL and WCF; certified root: RPZ1-19. Full human citations accompany the proof.

Figure 1: The two parameter branches are the exact coordinate axes of RLA6-10; their coupled module, rather than an uncoupled sum, is RLA14-18. These are schematic local germs, not a metric drawing or a certified numerical neighborhood. The right chart is the actual real parameter slice through the exact inverse RLA9. A collision of nonzero spectral roots does not make g_0 vanish. RQT3-16 constructs the bottom maps with the original Gamma metric: T_0 maps to orthonormal quotient coordinates q , while the alternative coordinate $w = M^{-1/2}q$ has Gram M . Both angle sines and their full volume correction are computed exactly. Original family and metric: [3, 10]; certified root: [9]; complete directional antecedent: [6].

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